

Sec. A (Attempt any two questions) 6 x 2

1. Find the n th derivative of $\sinh 2x \sin 4x$.
2. Find the radius of curvature of the cardioid $r = a(1 + \cos \theta)$ at any point (r, θ) on it. Prove that ρ^2/r is a constant.
3. Evaluate $\lim_{x \rightarrow \pi/4} \frac{\sec^2 x - 2 \tan x}{1 + \cos 4x}$
4. Expand $\log x$ in powers of $(x-1)$ and hence find the value of $\log_e 1.1$.

Sec. B (Attempt any one question) 8 x 1

1. If $y = (x^2 - 1)^n$, prove that $(x^2 - 1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$
2. Find all the asymptotes of the curve $4x^3 - 3xy^2 - y^3 + 2x^2 - xy - y^2 - 1 = 0$

Sec. A (Attempt any two questions) 6 x 2

1. If $u = z - x$, $v = y - z$, $w = x + y + z$, find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.
2. Examine the function $x^2 + 2y^2 - 4x + 4y + 6$ for extreme.
3. Define the rank of a matrix and find the rank of the matrix $A = \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \end{bmatrix}$.
4. Expand x^2y about the point $(1, -2)$.

Sec. B (Attempt any one question) 8 x 1

1. Test for consistency and solve $x + 2y + 3z = 6$,
 $2x + y - z = -3$, $3x - y + 2z = 11$

2. If $u = \sin^{-1} \left[\frac{x^{1/4} + y^{1/4}}{x^{1/6} + y^{1/6}} \right]$ then evaluate

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$$

$z - x - u$
 $y - z - v$
 $x + y + z - w$